

|                                                     |                                                                |                                                                |                                                     |                                           |                                |                                |
|-----------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------|--------------------------------|--------------------------------|
|                                                     | $\mathcal{K}_0^{\#1}$                                          | $\mathcal{K}_0^{\#1}{}_{\alpha\beta\chi}$                      |                                                     |                                           |                                |                                |
| $\mathcal{K}_0^{\#1}{}^\dagger$                     | $-\frac{\Upsilon_1 k^2}{4}$                                    | 0                                                              |                                                     |                                           |                                |                                |
| $\mathcal{K}_0^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                                              | $\Upsilon_2 k^2 + 3 \overset{(2)}{\kappa}_1$                   | $\mathcal{K}_1^{\#1}{}_{\alpha\beta}$               | $\mathcal{K}_1^{\#2}{}_{\alpha\beta}$     | $\mathcal{K}_1^{\#1}{}_\alpha$ | $\mathcal{K}_1^{\#2}{}_\alpha$ |
| $\mathcal{K}_1^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{1}{2}(\Upsilon_3 k^2 + 4 \overset{(2)}{\kappa}_1)$      | $\frac{\Upsilon_3 k^2 + 4 \overset{(2)}{\kappa}_1}{2\sqrt{2}}$ | 0                                                   | 0                                         |                                |                                |
| $\mathcal{K}_1^{\#2}{}^\dagger{}^{\alpha\beta}$     | $\frac{\Upsilon_3 k^2 + 4 \overset{(2)}{\kappa}_1}{2\sqrt{2}}$ | $\frac{1}{4}(\Upsilon_3 k^2 + 4 \overset{(2)}{\kappa}_1)$      | 0                                                   | 0                                         |                                |                                |
| $\mathcal{K}_1^{\#1}{}^\dagger{}^\alpha$            | 0                                                              | 0                                                              | 0                                                   | 0                                         |                                |                                |
| $\mathcal{K}_1^{\#2}{}^\dagger{}^\alpha$            | 0                                                              | 0                                                              | 0                                                   | 0                                         |                                |                                |
|                                                     |                                                                |                                                                | $\mathcal{K}_2^{\#1}{}_{\alpha\beta}$               | $\mathcal{K}_2^{\#1}{}_{\alpha\beta\chi}$ |                                |                                |
|                                                     |                                                                |                                                                | $\mathcal{K}_2^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{\Upsilon_4 k^2}{2}$                | 0                              |                                |
|                                                     |                                                                |                                                                | $\mathcal{K}_2^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                         | $\frac{\Upsilon_4 k^2}{2}$     |                                |

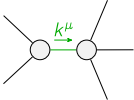
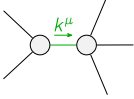
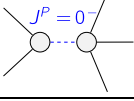
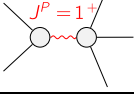
|                                                         |                                     |                                               |                                                         |                                           |                                           |                                               |  |
|---------------------------------------------------------|-------------------------------------|-----------------------------------------------|---------------------------------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------------------|--|
|                                                         | $\mathcal{T}_{0^+}^{\#1}$           | $\mathcal{T}_{0^+}^{\#1}{}_{\alpha\beta\chi}$ |                                                         |                                           |                                           |                                               |  |
| $\mathcal{T}_{0^+}^{\#1}{}^\dagger$                     | $\frac{1}{\text{Det}(0^+)}$         | 0                                             |                                                         |                                           |                                           |                                               |  |
| $\mathcal{T}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                   | $\frac{1}{\text{Det}(0^+)}$                   | $\mathcal{T}_{1^+}^{\#1}{}_{\alpha\beta}$               | $\mathcal{T}_{1^+}^{\#2}{}_{\alpha\beta}$ | $\mathcal{T}_{1^+}^{\#1}{}_\alpha$        | $\mathcal{T}_{1^+}^{\#2}{}_\alpha$            |  |
| $\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{2}{3\text{Det}(1^+)}$        | $\frac{\sqrt{2}}{3\text{Det}(1^+)}$           | 0                                                       | 0                                         |                                           |                                               |  |
| $\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$     | $\frac{\sqrt{2}}{3\text{Det}(1^+)}$ | $\frac{1}{3\text{Det}(1^+)}$                  | 0                                                       | 0                                         |                                           |                                               |  |
| $\mathcal{T}_{1^+}^{\#1}{}^\dagger{}^\alpha$            | 0                                   | 0                                             | 0                                                       | 0                                         |                                           |                                               |  |
| $\mathcal{T}_{1^+}^{\#2}{}^\dagger{}^\alpha$            | 0                                   | 0                                             | 0                                                       | 0                                         |                                           |                                               |  |
|                                                         |                                     |                                               |                                                         |                                           | $\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta}$ | $\mathcal{T}_{2^+}^{\#1}{}_{\alpha\beta\chi}$ |  |
|                                                         |                                     |                                               | $\mathcal{T}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$     | $\frac{1}{\text{Det}(2^+)}$               | 0                                         |                                               |  |
|                                                         |                                     |                                               | $\mathcal{T}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$ | 0                                         | $\frac{1}{\text{Det}(2^+)}$               |                                               |  |

#### Abbreviations used in matrices

$$\begin{aligned} \Upsilon_1 &== 6 \overset{(4)}{\kappa}_1 + 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \&\& \Upsilon_2 == \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} \&\& \Upsilon_3 == 2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - 2 \overset{(4)}{\kappa}_3 \&\& \\ \Upsilon_4 &== 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} \&\& \text{Det}(0^+) == -\frac{1}{4}k^2(6 \overset{(4)}{\kappa}_1 + 2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}) \&\& \\ \text{Det}(1^+) &== \frac{3}{4}k^2(2 \overset{(4)}{\kappa}_{15} - \overset{(4)}{\kappa}_{16} - 2 \overset{(4)}{\kappa}_3) + 3 \overset{(2)}{\kappa}_1 \&\& \text{Det}(2^+) == \frac{1}{2}k^2(2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}) \end{aligned}$$

#### Lagrangian

$$\begin{aligned} &\overset{(2)}{\kappa}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{\kappa}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^\delta_{\chi\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \overset{(4)}{\kappa}_{15} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \frac{1}{2} \overset{(4)}{\kappa}_{16} \partial_\chi \mathcal{K}^\delta_{\beta\delta} \partial^\chi \mathcal{K}^\alpha{}_\beta + \\ &\overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\alpha\chi} + 2 \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \overset{(4)}{\kappa}_{16} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - 2 \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \\ &\overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\chi\alpha} - 2 \overset{(4)}{\kappa}_{15} \partial^\chi \mathcal{K}^\alpha{}_\beta \partial_\delta \mathcal{K}^\delta_{\chi\beta} - \overset{(4)}{\kappa}_{16} \partial^\chi \mathcal{K}^\alpha{}_\beta \partial_\delta \mathcal{K}^\delta_{\chi\beta} - \frac{1}{2} \overset{(4)}{\kappa}_{15} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \\ &\frac{1}{4} \overset{(4)}{\kappa}_{16} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} - \frac{1}{2} \overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta_{\beta\chi} + \overset{(4)}{\kappa}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \overset{(4)}{\kappa}_{16} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

| Added source term(s):                                                              | $\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$ |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                     |
|------------------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Source constraint(s)                                                               | # constraint(s)                                               | Covariant form                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                     |
| $\mathcal{J}_1^{\#2\alpha} == 0$                                                   | 3                                                             | $\partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$                                                                                                                                                                                                                                                                                                                                            |                                                                                                     |
| $\mathcal{J}_1^{\#1\alpha} == 0$                                                   | 3                                                             | $\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$                                                                                                                                                                                                               |                                                                                                     |
| $\mathcal{J}_1^{\#1\alpha\beta} == \mathcal{J}_1^{\#2\alpha\beta}$                 | 3                                                             | $3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$ |                                                                                                     |
| Total # constraint(s):                                                             | 9                                                             |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                     |
| Resolved pole(s)                                                                   | # polarization(s)                                             | Square mass                                                                                                                                                                                                                                                                                                                                                                                                  | Residue                                                                                             |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                            | $\frac{1}{2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$                               |
|  | 2                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                            | $-\frac{1}{2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$                              |
|  | 1                                                             | $\frac{3 \overset{(2)}{\kappa}_1}{-\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$                                                                                                                                                                                                                                                                                                                 | $\frac{1}{-\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16}}$                                |
|  | 3                                                             | $\frac{4 \overset{(2)}{\kappa}_1}{-2 \overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{16} + 2 \overset{(4)}{\kappa}_3}$                                                                                                                                                                                                                                                                                   | $\frac{4}{6 \overset{(4)}{\kappa}_{15} - 3 \overset{(4)}{\kappa}_{16} + 2 \overset{(4)}{\kappa}_3}$ |
| Resolved unitarity condition(s):                                                   |                                                               | (Demonstrably impossible)                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                     |